

Project Design Phase
Proposed Solution Template

Date	27 June 2026
Team ID	SWTID-2026-5525
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	2 Marks

Proposed Solution Template:

S.No	Parameter	Description
1	Problem Statement (Problem to be solved)	Electric vehicle information such as price, range, efficiency, charging station availability and brand performance is available across different sources, making it difficult for users and analysts to compare and understand EV market insights effectively.
2	Idea / Solution description	Develop an Electric Cars Analytics Platform that collects EV datasets, stores data in MySQL, performs SQL-based analysis and creates interactive Tableau dashboards. The dashboards are integrated into a Flask web application to provide easy access to EV insights.
3	Novelty / Uniqueness	The solution combines database management, data analytics, visualization and web integration into a single platform. It provides interactive filters, EV comparison, charging station analysis, brand performance insights and user-friendly dashboards.
4	Social Impact / Customer Satisfaction	Helps customers make better EV purchasing decisions by understanding price, performance, efficiency and charging availability. It supports the adoption of electric vehicles through transparent data-driven insights.
5	Business Model (Revenue Model)	The platform can provide subscription-based EV analytics services for automobile companies, researchers and businesses. It can also generate revenue through premium analytics reports and EV market insights.
6	Scalability of the Solution	The system can be expanded by adding more EV datasets, real-time charging station information, predictive analytics, AI-based recommendations and support for global EV markets.